

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Internship: Data Engineering Internship

Week: Week 4 Assignment

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Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

1 Environment Setup & Storage Provisioning

1.1 Task 1: Azure Portal & Resource Group

1. Logged into Microsoft Azure portal.
2. Opened Resource Groups and clicked Create.
3. Entered resource group name as rg-celebal-week4.
4. Selected region as East US.
5. Created the resource group successfully.

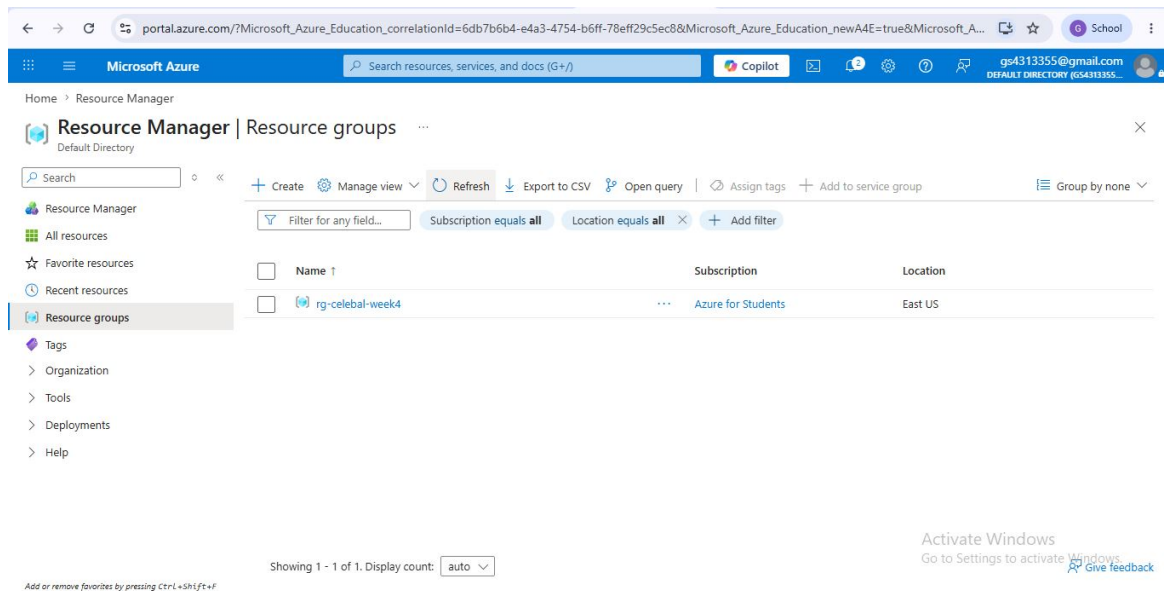


Figure 1: : Resource Group Creation

1.2 Task 2: Storage Setup

1. Opened Storage Accounts and clicked Create.
2. Entered storage account name as stcelebalweek4.
3. Selected Performance = Standard.

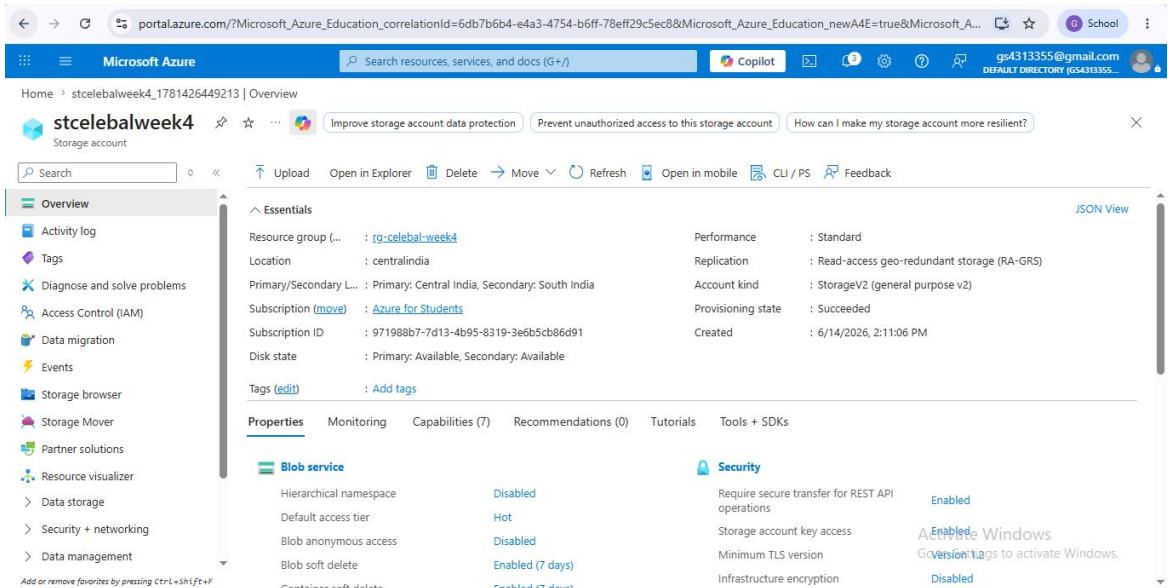


Figure 2: : Storage Account Creation

4. Created a blob container named `raw-data` inside the storage account.
5. Uploaded `Sample - Superstore.csv` into the container successfully.

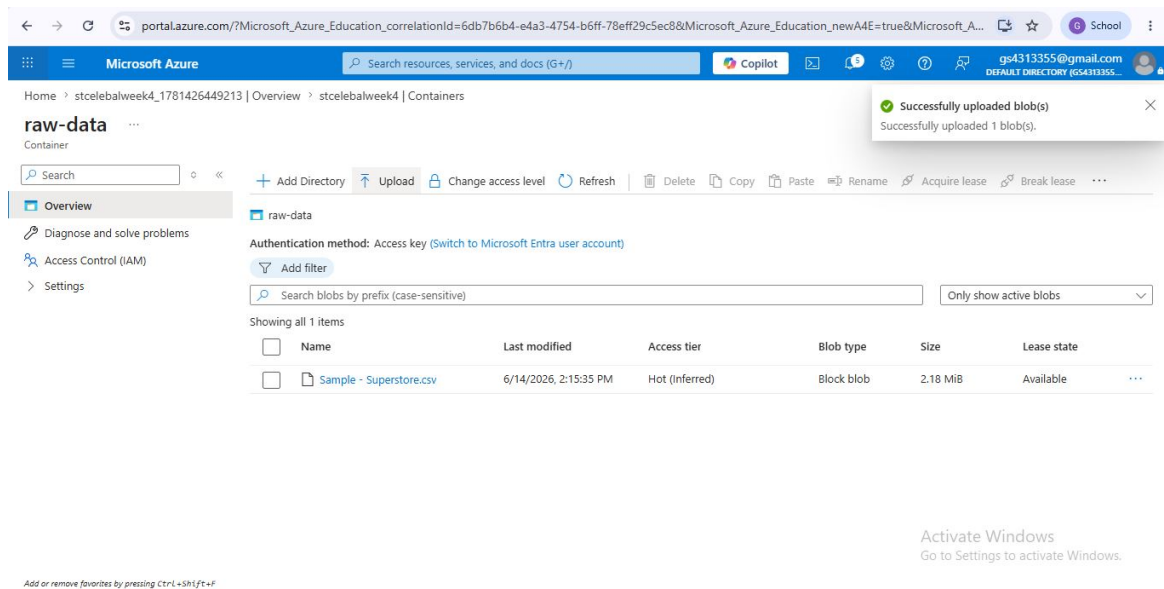


Figure 3: : Container with Uploaded CSV

2 Data Factory Deployment & Connectors

2.1 Task 3: ADF Basics

1. Opened Data Factories in the Azure Portal.
2. Verified the creation and status of the data factory named `adf-celebal-week4`.

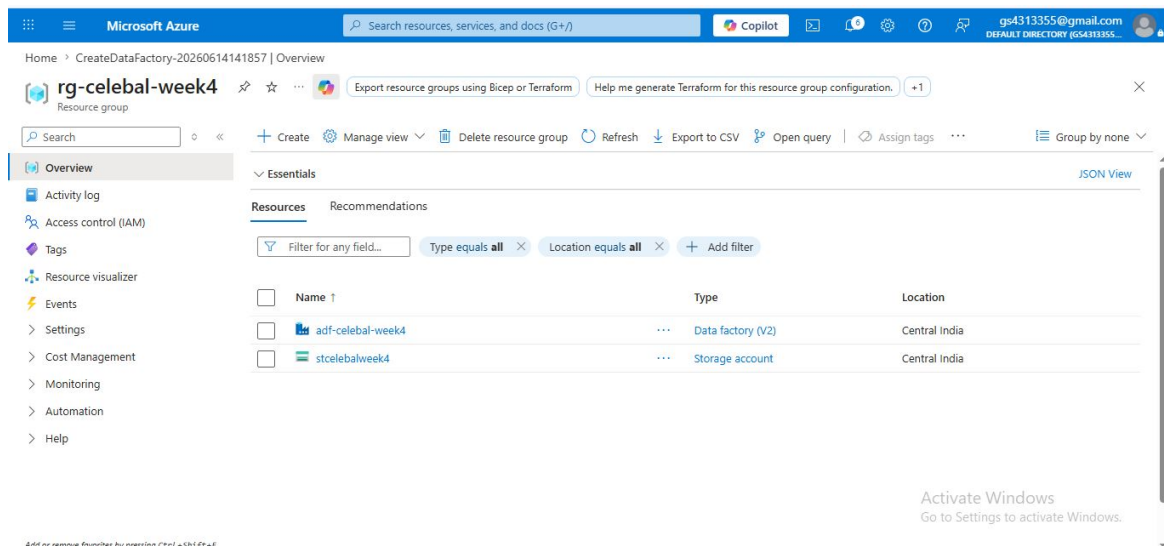
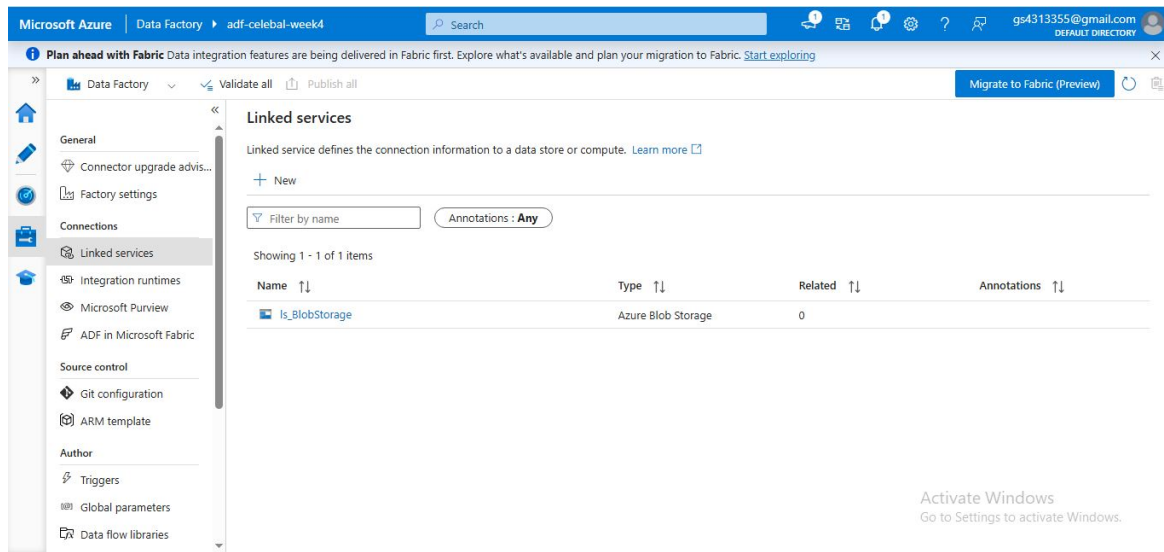
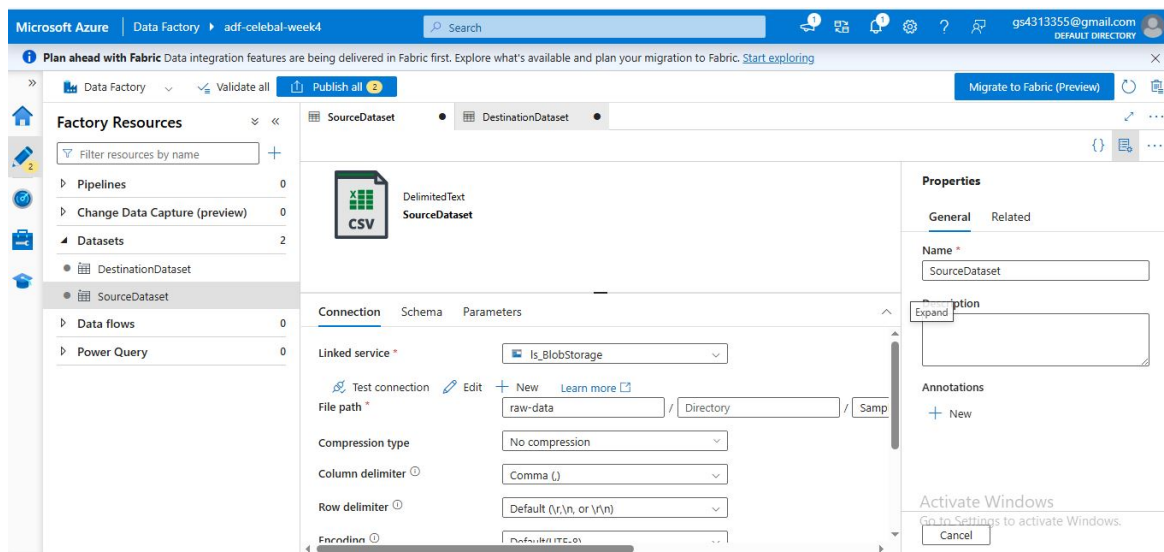


Figure 4: : Data Factory Provisioning

3. Opened ADF Studio, navigated to Manage → Linked Services, and clicked New.
4. Selected Azure Blob Storage and created a linked service named `ls_BlobStorage`.
5. Successfully verified the connection to the backend storage account.

Figure 5: : *Linked Service Connection Successful*

6. Navigated to the Author tab and created a new source dataset named `Sourcedataset`.
7. Configured the format as `DelimitedText (CSV)` pointing straight to the file location.

Figure 6: : *Source Dataset Object Setup*

8. Created a new structural destination dataset reference named DestinationDataset.
9. Formed a clean processing data integration pipeline canvas named pl_process_superstore.
10. Dragged a Get Metadata activity (Get Metadata1) onto the canvas grid window.
11. Configured its properties mapping details to point directly to Sourcedataset.

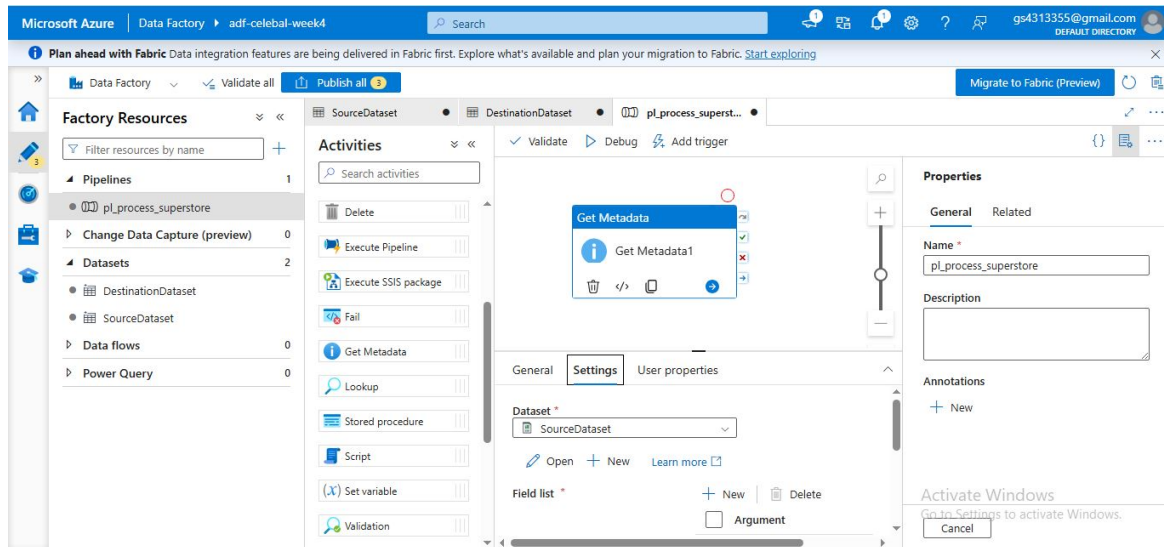


Figure 7: : Get Metadata Activity Configuration

2.2 Task 4: Pipeline Development

1. Dragged a Copy Data activity (Copy data1) onto the pipeline workflow canvas.
2. Linked the green on-success activity connector path from the metadata block to the Copy block.
3. Configured the internal properties to use Sourcedataset as source and DestinationDataset as sink.

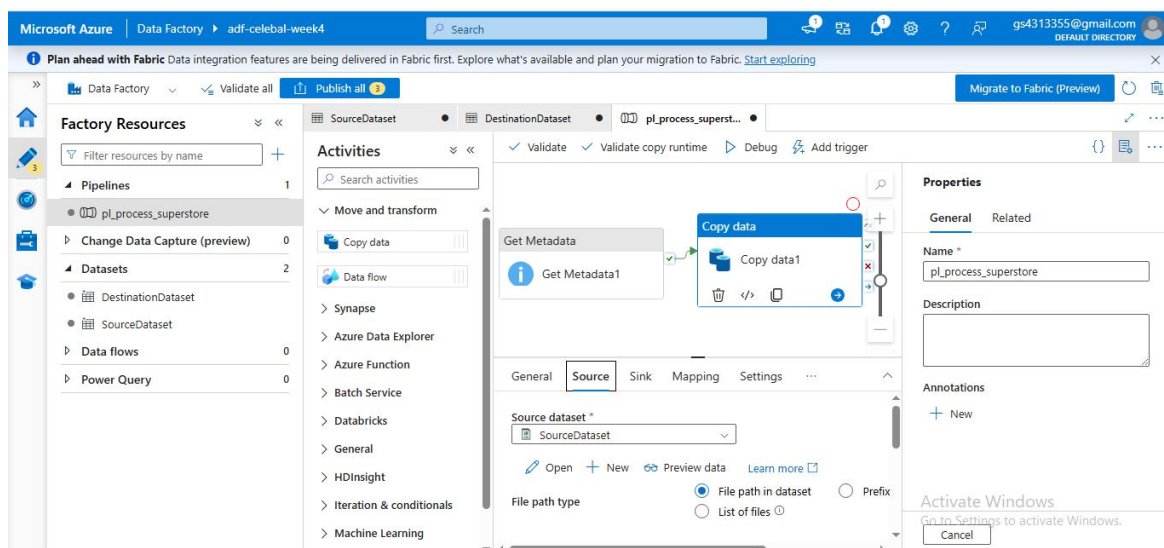


Figure 8: : End-to-End Pipeline Logical Flow

3 Orchestration Run Tracking & Governance

3.1 Task 5: Pipeline Execution

1. Validated pipeline structure and triggered an execution cycle using the **Debug** runner.
2. Tracked runtime status changes inside the output monitor sub-drawer tracking tray.
3. Verified execution logs showing that both activities successfully completed with a status of **Succeeded**.

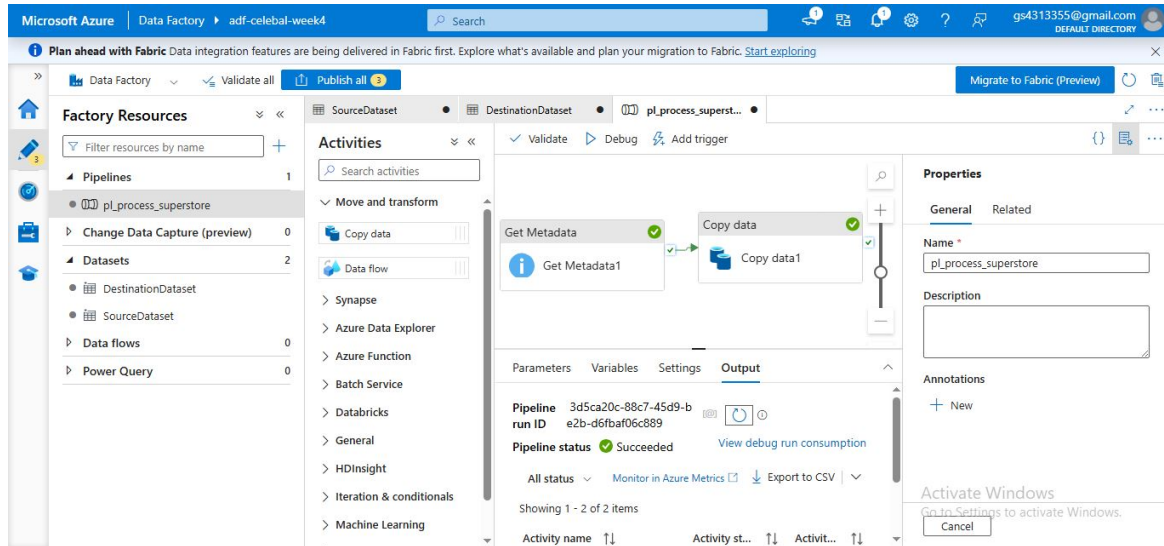


Figure 9: : Pipeline Execution Status Monitor

3.2 Task 6: IAM Roles

1. Opened the stcelebalweek4 Storage Account workspace dashboard tier view.
2. Navigated into the left-hand navigation option panel for **Access Control (IAM)**.
3. Inspected the security group assignments and active access matrix parameters for resources.

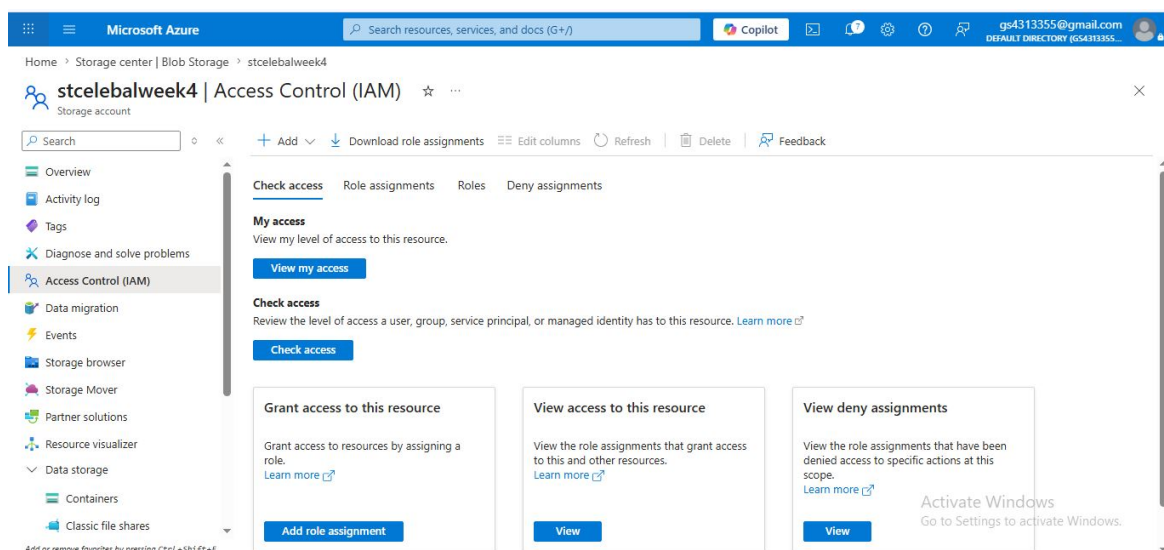


Figure 10: : Access Control (IAM) Matrix Panel Interface